

Hyeongmin Lee

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Research Interest

I'm interested in all the phenomena that happen in videos. I like to explore and use them to create valuable technologies.

- Machine Learning & Computer Vision
- Video Understanding & Representation Learning
- Video Processing & Enhancement & Compression
- 3D Vision
- World Models

Education

Mar.2018 – Aug.2023 **INTEGRATED M.S./PH.D. Course in Electrical and Electronic Engineering**
Yonsei University, Seoul, Korea

- Image and Video Pattern Recognition LAB
- Supervised by Prof. Sangyoun Lee
- GPA: 4.24/4.3

Mar.2014 – Feb.2018 **B.S. in Electrical and Electronic Engineering**
Yonsei University, Seoul, Korea

- Graduated **Cum Laude**
- GPA: 3.96/4.3

Work Experience

Mar.2026 - Present **Seoul National University of Science and Technology (SeoulTech)**, Seoul, Korea
Assistant Professor

- Department of Electronic Engineering

Jul.2023 - Feb.2026 **Twelve Labs**, Seoul, Korea
ML Research Scientist

- Working on the Foundation Models for Video Understanding

May.2021 - Nov.2021 **Adobe Research**, San Jose, CA (Remote)
Research Intern

- Working with Dr. Simon Niklaus and Dr. Oliver Wang
- Research on Video Stabilization

Teaching

- Engineering Mathematics 1 (2026-1)
- Image Processing (2026-1)

Publications

- 2026 [1] **Hyeongmin Lee** and Kyungjune Baek, “Probing Intrinsic Bias: Internal Attention Feature Analysis for Social Bias Evaluation in Diffusion Models,” *Neurocomputing*, 2026.
- 2025 [2] **Hyeongmin Lee** and Kyungjune Baek, “Temporal Smoothness-Aware Rate-Distortion Optimized 4D Gaussian Splatting,” *Neural Information Processing Systems (NeurIPS)*, 2025.
- 2024 [3] Raehyuk Jung*, Hyojun Go*, Jaehyuk Yi*, Jiho Jang*, Daniel Kim*, and Jay Suh* et al., “Pegasus-v1 Technical Report,” *Twelve Labs Technical Report*, 2024.
- [4] **Hyeongmin Lee***, Jin-Young Kim*, Kyungjune Baek*, and Jihwan Kim* et al., “TWLV-I: Analysis and Insights from Holistic Evaluation on Video Foundation Models,” *Twelve Labs Technical Report*, 2024.
- 2023 [5] **Hyeongmin Lee**, Taeoh Kim, Hanbin Son, Sangwook Baek, Minsu Cheon, and Sangyoun Lee, “A Nonlinear, Regularized, and Data-Independent Modulation for Continuously Interactive Image Processing Network,” *International Journal of Computer Vision*, 2023.
- [6] Sangjin Lee*, **Hyeongmin Lee***, Chajin Shin, and Sangyoun Lee, “Exploring Discontinuity for Video Frame Interpolation,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (Highlight; Top 10% of Accepted Papers)*, 2023.
- 2022 [7] Chajin Shin, **Hyeongmin Lee**, Hanbin Son, Sangjin Lee, Dogyoon Lee, and Sangyoun Lee, “Expanded Adaptive Scaling Normalization for End to End Image Compression,” *European Conference on Computer Vision (ECCV)*, 2022.
- [8] Hanbin Son, Taeoh Kim, **Hyeongmin Lee**, and Sangyoun Lee, “Enhanced Standard Compatible Image Compression Framework based on Auxiliary Codec Networks,” *IEEE Transactions on Image Processing*, 2022.
- 2021 [9] Dogyoon Lee, Jaeha Lee, Junhyeop Lee, **Hyeongmin Lee**, Minhyeok Lee, Sungmin Woo, and Sangyoun Lee, “Regularization Strategy for Point Cloud via Rigidly Mixed Sample,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- 2020 [10] Taeoh Kim*, **Hyeongmin Lee***, MyeongAh Cho*, Hoseong Lee, Dong heon Cho, and Sangyoun Lee, “Learning Temporally Invariant and Localizable Features via Data Augmentation for Video Recognition,” *European Conference on Computer Vision Workshops (ECCVW) (Oral)*, 2020.
- [11] **Hyeongmin Lee**, Taeoh Kim, Tae-young Chung, Daehyun Pak, Yuseok Ban, and Sangyoun Lee, “AdaCoF: Adaptive Collaboration of Flows for Video Frame Interpolation,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [12] Sangjin Lee, **Hyeongmin Lee**, Taeoh Kim, and Sangyoun Lee, “Extrapolative-Interpolative Cycle-Consistency Learning for Video Frame Extrapolation,” *IEEE International Conference on Image Processing (ICIP)*, 2020.
- 2019 [13] MyeongAh Cho, Tae-young Chung, **Hyeongmin Lee**, and Sangyoun Lee, “N-RPN : Hard Example Learning for Region Proposal Networks,” *IEEE International Conference on Image Processing (ICIP)*, 2019.
- [14] Taeoh Kim, **Hyeongmin Lee**, Hanbin Son, and Sangyoun Lee, “SF-CNN : A Fast Compression Artifacts Removal via Spatial-to-Frequency Convolutional Neural Networks,” *IEEE International Conference on Image Processing (ICIP)*, 2019.

- 2018 [15] **Hyeongmin Lee**, Taeoh Kim, Eungyeol Song, and Sangyoun Lee, “CollaboNet: Collaboration of Generative Models by Unsupervised Classification,” *IEEE International Conference on Image Processing (ICIP)*, 2018.
- [16] Eungyeol Song, **Hyeongmin Lee**, Jaesung Choi, and Sangyoun Lee, “AHD: Thermal Image-based Adaptive Hand Detection for Enhanced Tracking System,” *IEEE Access*, 2018.

* indicates equal contribution. Click the titles to move to the paper links

Patents

- 2023 **Apparatus and Method for Selecting Skip Frames**
Korea Patent no.10-2620281, Dec. 2023
- 2021 **Apparatus for compressing image, learning apparatus and method thereof**
Korea Patent no.10-2245682, Apr. 2021
- 2021 **Image processing apparatus and operating method for the same**
Korea Patent no.10-2236582, Mar. 2021
- 2021 **Apparatus and method for interpolating frames based on multiple flows**
Korea Patent no.10-2201297, Jan. 2021

Projects

- Sep.2025 – Feb.2026 **Video Vertical Voyage: Semantic Chunking and Hierarchical Pre-merging**
Twelve Labs
- May.2025 – Sep.2025 **Video Moment Retrieval**
Twelve Labs
- Aug.2024 – Jan.2025 **Video Embedding Vector Quantization**
Twelve Labs
- Jul.2023 – Aug.2024 **TWLV-I: Analysis and Insights from Holistic Evaluation on Video Foundation Models**
Twelve Labs
- Jul.2022 – Jun.2023 **Development of Enhanced Deep Video Compression Framework**
Samsung Research
- Jun.2021 – May.2022 **Development of Deep Learning-based Video Frame Interpolation Algorithm for Game and AR Videos**
Samsung Electronics Mobile Division
- Jun.2021 – May.2022 **Deep Learning-based E2E Video Codec Research**
Samsung Research
- May.2021 – Nov.2021 **Fast Video Stabilization**
Adobe Research
- Apr.2020 – Dec.2020 **Development of Data Augmentation-based AI Algorithm for Image Enhancement**
Samsung Research

May.2019 – Dec.2019	Locally Controllable Deep Neural Network Algorithm Development <i>Samsung Research</i>
Apr.2018 – Dec.2018	Development of Compact Deep Networks for Image Processing <i>Samsung Research</i>
Apr.2017 – Dec.2017	Broadcasting Device Controlling & Editing Technology using Personal Voice/Gesture Recognition <i>Institute for Information & Communications Technology Promotion (IITP), Ministry of Science</i>

Honors & Awards

- 2021 **5th Next Generation Engineering Leader Award of NAEK**
National Academy of Engineering of Korea (NAEK)
- 2020 **Visual Inductive Priors for Data-Efficient Action Recognition Challenge**
1st Visual Inductive Priors for Data-Efficient Deep Learning Workshop
 - European Conference on Computer Vision (ECCV) Workshop
 - **4th Place**
- 2018 **Honor Graduation Award**
Electrical & Electronic Engineering, Yonsei University
 - Cum Laude
- 2014 - 2017 **Honor Student Award**
Electrical & Electronic Engineering, Yonsei University
 - The Highest Honor Student Award (2014-1)
 - Honor Student Award (2014-2, 2015-1, 2015-2, 2017-1, 2017-2)

Invited Talks

- **AdaCoF: Adaptive Collaboration of Flows for Video Frame Interpolation**
– GPU Technology Conference (GTC), *NVIDIA*, Oct.2020
- **Introduction to Image Translation**
– Junior Seminar, *Electrical & Electronic Engineering, Yonsei University*, Nov.2020
- **Complex and discontinuous motion compensation for video frame interpolation**
– Multimodal Weekly Webinar, *Twelve Labs*, Feb.2024
- **Video Foundation Models**
– THE AI KOREA 2024, *KIISE Artificial Intelligence Society*, Aug.2024

Peer Reviews

- IEEE Open Journal of Signal Processing (2021)
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2022, 2023, 2024, 2025)
- European Conference on Computer Vision (ECCV) (2022, 2024)
- AAAI Conference on Artificial Intelligence (AAAI) (2025)
- Conference on Neural Information Processing Systems (NeurIPS) (2025)
- The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (2026)
- ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM) (2025)

Activities

Mar.2018 – Dec.2018 **Teaching Assistant**

Electrical and Electronic Engineering, Yonsei University

- (2018-1) Digital Signal Processing
- (2018-2) Signal and System

Nov.2016 – Present **Young Engineers' Honor Society (YEHS)**

Organization under The National Academy of Engineering of Korea (NAEK)

- Deputy Department Head of Academic Exchange Dpt. (Dec.2017 – Dec.2018)
- Department Head of Academic Exchange Dpt. (Dec.2018 – Dec.2019)
- President (Dec.2019 – Dec.2020)
- Standing Commissioner (Dec.2020 - Present)

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